

Practice Quiz: Planning -- Simulating Futures to Act Now

Part A: Multiple Choice

1. Planning in Active Inference is best described as: A) Making a to-do list B) Mentally simulating multiple possible futures and selecting the action policy that minimizes expected free energy C) Predicting exactly what will happen D) Avoiding all uncertain situations

Answer: B. Planning uses your generative model to simulate future scenarios and evaluate which path leads to the lowest overall predicted surprise.

1. When planning your weekend schedule, you mentally simulate studying Saturday morning versus Sunday night. This involves: A) Guessing randomly B) Your generative model evaluating each scenario's expected free energy -- factoring in tiredness, missed fun, and exam stress C) Flipping a coin D) Doing whatever your friends do

Answer: B. Your model weighs pragmatic and epistemic factors across different timelines to select the best policy.

1. Organizing a birthday party involves Active Inference planning because: A) Parties do not require planning B) You build a generative model of what could go wrong and take preemptive actions to reduce the expected free energy of each contingency C) You only need to buy a cake D) Planning guarantees nothing will go wrong

Answer: B. Ordering extra food, having an indoor backup, and anticipating guest behavior all reduce expected surprise.

1. Good planners differ from poor planners primarily in: A) How much time they spend worrying B) The depth and accuracy of their generative models for simulating future consequences of current actions C) How many planners they own D) Whether they follow rules

Answer: B. Better generative models produce more accurate simulations, leading to policies with lower expected free energy.

1. **Procrastination** can be understood in Active Inference as: A) Laziness B) Selecting a policy that minimizes short-term free energy (avoiding the discomfort of uncertainty) while increasing long-term expected free energy (mounting deadline stress) C) Good time management D) A sign of high intelligence

Answer: B. The procrastinator's model prioritizes immediate comfort over future consequences -- a short-sighted policy evaluation.

1. A backup plan reduces expected free energy because: A) It means you expect to fail B) It provides an alternative action policy that handles contingencies, reducing the overall uncertainty about future outcomes C) Backup plans are never needed D) It eliminates all risk

Answer: B. Contingency planning expands your policy space so that more future scenarios have an adequate response.

1. Why does planning for college applications feel overwhelming to many high schoolers? A) College is not important B) The generative model must simulate many uncertain variables simultaneously (acceptance odds, financial aid, fit, future career paths), creating high expected free energy C) Applications are too easy D) Only anxious people find it overwhelming

Answer: B. The planning horizon is long, the variables are numerous, and many outcomes are genuinely uncertain.

1. Setting a specific goal (like "run a 5K by June") improves planning because: A) Goals are always achieved B) A concrete goal provides a clear target state against which your generative model can evaluate action policies and measure progress C) Vague goals work equally well D) Goals eliminate all prediction error

Answer: B. Specific goals reduce ambiguity in the expected free energy calculation, making it easier to evaluate which actions move you closer.

1. Re-planning when circumstances change (like finding out a study group is cancelled) demonstrates: A) That the original plan was bad B) Flexible Active Inference -- updating your action policy when new evidence changes the expected free energy landscape C) That planning is useless D) Rigid thinking

Answer: B. Good planning is not about sticking to one plan but about continuously re-evaluating policies as new information arrives.

1. Imagining the worst-case scenario can sometimes **help** planning because: A) Pessimism is always healthy B) It forces your generative model to simulate high-surprise futures and develop contingency policies that reduce their expected free energy C) Worst cases always happen D) It increases anxiety, which improves performance

Answer: B. "Pre-mortem" thinking expands the range of futures your model considers, leading to more robust plans.

Part B: Short Answer

1. Describe a plan you made that had to be revised when circumstances changed. How did the new information update the expected free energy of your original policy?
2. Explain why breaking a large goal (like "get into college") into smaller sub-goals (like "write essay draft by Friday") makes planning more effective in Active Inference terms.
3. How does procrastination create a trap where the cost of starting keeps increasing? Use the concepts of short-term versus long-term expected free energy to explain.
4. Compare planning a surprise party (where you control many variables) versus planning for a job interview (where many variables are outside your control). How does the difference in uncertainty affect your planning strategy?

5. Design a one-week study plan for final exams using Active Inference principles. How would you balance exploration (studying weak areas) with exploitation (reinforcing strong areas)?